REQUIREMENTS GATHERING

Store Sales Dataset Analysis

Phase 1 | Planning and Requirements

Purpose
Define the project stakeholders, user stories and use cases, functional requirements, and non-functional requirements for the Store Sales Dataset Analysis project.

Dataset fields
Category, City, Country, Customer ID, Customer Name, Order Date, Order ID, Postal Code, Product ID, Product Name, Region, Row ID, Sales, Segment, Ship Date, Ship Mode, State, and Sub-Category.

1. Introduction

The project will prepare the available sales data, answer business questions, build an interactive Power BI dashboard, and deliver a final report and presentation. The analysis is based on the dataset fields listed above.

2. Stakeholder Analysis

Stakeholders are people who use, evaluate, or build the analysis. Influence and interest are planning estimates.

Stakeholder	Role	Needs and Expectations	Influence / Interest
Store / Business Manager	Primary user of insights	Clear sales performance; top and bottom products, categories, and regions.	High / High
Sales and Marketing Team	Plans sales actions	Sales trends and products, categories, or segments contributing most to sales.	Medium / High
Inventory and Operations Team	Operational planning	Product sales patterns and shipping modes to support operational decisions.	Medium / Medium
Senior Management	Reviews summaries	Concise KPIs and findings without technical detail.	High / Medium
Course Instructor	Evaluates the project	Complete documentation, sound analysis, working dashboard, and clear presentation.	High / High
Project Team / Data Analysts	Builds the solution	Agreed requirements, usable dataset, clear tasks, and realistic plan.	High / High

3. User Stories and Use Cases

3.1 User Stories

User stories describe what each user wants from the solution and why.

ID	As a...	I want to...	So that...
US-01	Business Manager	See total sales and order volume on one dashboard page	Understand overall sales performance.
US-02	Business Manager	Compare sales by category and region	Identify strongest and weakest areas.
US-03	Sales / Marketing	View monthly and yearly sales trends	Spot patterns and plan campaigns.
US-04	Sales / Marketing	Filter by date, region, and category	Focus on the relevant business area.
US-05	Operations Team	See top- and bottom-selling products	Support operational decisions.
US-06	Senior Manager	Read a short summary of KPIs and findings	Understand results without technical detail.
US-07	Data Analyst	Identify missing values, duplicates, and incorrect data types	Use reliable data.
US-08	Data Analyst	Answer business questions using available fields	Deliver evidence-based insights.
US-09	Instructor	Review documentation, dashboard, and final report	Evaluate project deliverables

3.2 Use Cases

The main actors are the Data Analyst, Business User, and Instructor.

ID	Use Case	Primary Actor	Description
UC-01	Import Sales Dataset	Data Analyst	Load the available dataset into the analysis environment.
UC-02	Clean and Prepare Data	Data Analyst	Inspect missing values, duplicates, formats, and data types; document unresolved issues.
UC-03	Analyze Sales Data	Data Analyst	Calculate sales KPIs and answer questions using available columns.
UC-04	Explore Dashboard	Business User	View KPI cards and charts and use filters to explore sales performance.
UC-05	Review Deliverables	Instructor	Review the report, dashboard, and presentation

4. Functional Requirements

Priority follows the MoSCoW scale: Must (essential), Should (important), and Could (desirable).

ID	Area	Requirement	Priority
FR-01	Data Management	Import the available sales dataset.	Must
FR-02	Data Management	Identify and handle missing values appropriately.	Must
FR-03	Data Management	Check duplicate records and document the treatment applied.	Must
FR-04	Data Management	Validate data types and formats, including dates and numeric Sales values.	Must
FR-05	Analysis	Calculate Total Sales, Total Orders, Average Order Value, and other KPIs supported by the available fields.	Must
FR-06	Analysis	Compare sales across products, sub-categories, categories, regions, and segments.	Must
FR-07	Analysis	Analyze monthly and yearly sales trends using Order Date.	Must
FR-08	Dashboard	Present key KPIs in summary cards on an overview page.	Must
FR-09	Dashboard	Include charts for sales by category, region, product, and time.	Must
FR-10	Dashboard	Provide filters for date, region, and category; segment may be included if useful.	Must
FR-11	Dashboard	Update visuals and KPI cards when applicable filters change.	Must
FR-12	Dashboard	Support navigation from summary views to product or trend details.	Should
FR-13	Reporting	Produce a report documenting methods, findings, limitations, and recommendations.	Must
FR-14	Reporting	Produce a final presentation summarizing results.	Must
FR-15	Reporting	Allow exporting relevant visuals or data where useful.	Could

KPIs supported by the listed fields

KPI	Calculation
Total Sales	Sum of Sales
Total Orders	Distinct count of Order ID
Average Order Value (AOV)	Total Sales divided by distinct Order ID count
Total Customers	Distinct count of Customer ID
Sales by Product / Category / Region / Segment	Sum of Sales grouped by the selected field
Sales Trend	Sum of Sales by month or year of Order Date
Average Shipping Days (optional)	Average days between Order Date and Ship Date, after validating date fields

5. Non-functional Requirements

These requirements define the quality attributes and constraints of the solution.

ID	Category	Requirement / Criteria
NFR-01	Performance	Dashboard interactions should update visuals within a few seconds.
NFR-02	Performance	Dataset loading and refresh should complete without errors or timeouts.
NFR-03	Security	Handle project data responsibly; avoid publishing customer-level details unnecessarily.
NFR-04	Access Control	Store project files with appropriate access permissions.
NFR-05	Usability	Use clear titles, labels, and a logical layout for non-technical users.
NFR-06	Visual Quality	Use appropriate chart types, readable fonts, and sufficient contrast.
NFR-07	Reliability	Validate KPI values against cleaned source data and totals/counts.
NFR-08	Reproducibility	Document cleaning and analysis so another member can repeat the work.
NFR-09	Maintainability	Use consistent naming and version-controlled project files.

6. Assumptions and Constraints

Assumptions

• The dataset listed in Power BI is the approved source.
• Power BI is used to build the dashboard.
• Team members have access to required software and the shared repository.
• KPIs are limited to fields available in the dataset.

Constraints

• The project must be completed within the assigned four-week period.
• Analysis is limited to information available in the dataset.
• All project documents must be uploaded to GitHub according to DEPI guidelines.